HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and validated **Python** programming code for statistical analysis and reporting on clinical trial data, ensuring quality and efficiency.
- Collaborated with **data scientists** and statisticians to define and implement data requirements for complex analytical models.
- Implemented **AI tools and technologies** within cloud environments to enhance clinical data analysis workflows.
- Managed programming practices to ensure compliance with **industry standards and regulatory guidelines** for clinical trial data processing.
- Created and maintained comprehensive **documentation** for programming processes, code repositories, and analytical outputs.
- Participated in the review of analysis datasets and final reports, ensuring accuracy and completeness for regulatory submissions.
- Deployed and managed **Python** applications and data pipelines on **AWS** services, including EC2 and Lambda, for scalable data processing.

SKILLS

Programming Languages – Python, SQL, Java

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

AI & Data Science – Machine Learning, Data Analysis, Statistical Programming, Knowledge Graphs, Data Engineering, Advanced Analytics

Tools & Technologies – Docker, Git, Jenkins, Jira, Confluence, Splunk, Dynatrace, PostgreSQL, MySQL, Oracle

Methodologies – Agile, Scrum, CI/CD

PROJECTS

Clinical Trial Data Analysis Platform

- Engineered a **Python**-based platform on **AWS** to process and analyze large volumes of clinical trial data, supporting statistical reporting needs.
- Developed data pipelines to ingest, clean, and transform raw clinical data, integrating with **PostgreSQL** for structured storage and retrieval.
- Implemented automated reporting modules using **Python** scripts, reducing manual report generation time by approximately 40% for study teams.

Knowledge Graph for Clinical Insights

- Designed and built a **knowledge graph** using **Python** to represent relationships between clinical trial parameters, patient demographics, and treatment outcomes.
- Extracted and structured data from diverse sources, including structured databases and free-text clinical notes, to populate the graph.
- Developed query interfaces to enable data scientists to explore complex relationships and uncover novel insights from the integrated data.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)